

HW 3, Problem 5

TFAE for $f(x) \in \mathbb{F}_p[x]$ irred.
of degree r .

(a) $r \mid d$

(b) $f(x) \mid x^{p^d} - x$.

Fact (1) $\mathbb{F}_p[x]$



$ev_\alpha: x \mapsto x + (f(x)) = \alpha$

$\frac{\mathbb{F}_p[x]}{\ker ev_\alpha}$

$\xrightarrow[\sim]{ev_\alpha}$

$\frac{\mathbb{F}_p[x]}{(f(x))} = k$



$\alpha = x + (f(x))$

Then

$\ker ev_\alpha = (f(x))$

Fact (2) : $|k| = p^r$

$\Rightarrow \alpha$ is a zero for

$$x^{p^r} - x \quad (\text{HW 2, Problem 5})$$

$$\text{i.e. } x^{p^r} - x \in \ker \text{ev}_\alpha = (f(x))$$

$$\Rightarrow \underline{f(x)} \mid x^{p^r} - x.$$

Rmk. old $\Rightarrow x^{p^r} - x \mid x^{p^d} - x$
 $\in \mathbb{Z}[x],$

[Needs to be shown]

Uniqueness part of Galois' theorem

(1) If k_1, k_2 are finite fields of order p^d , then

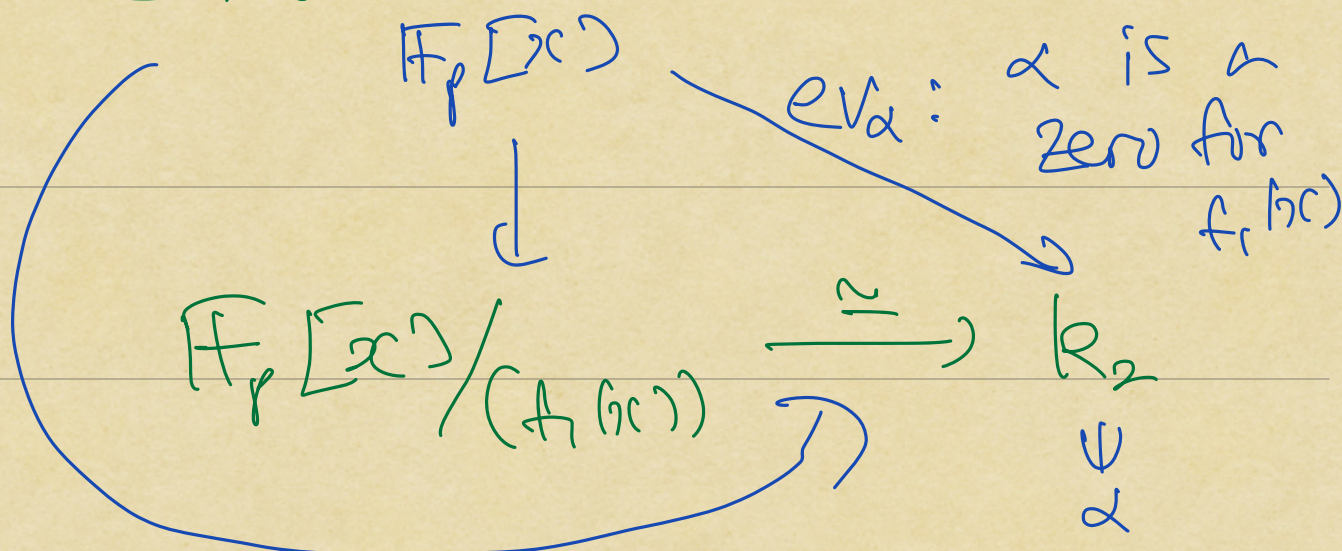
$$k_1 \simeq k_2$$

(2) If $k_1 = \mathbb{F}_p[x]/(f_1(x))$

$f_1(x) \in \mathbb{F}_p[x]$ irred. of deg d

then

$\exists$ isom



(3)

If $|k| = p^d$ and $f(x) \in \mathbb{F}_p[x]$ is irreducible of deg d , then $f(x)$ has a zero $\alpha \in k$.

Proposition:

If $|k| = p^d$ and $f(x) \in \mathbb{F}_p[x]$ irreducible of deg. $r \mid d$.

Then $f(x)$ has a zero $\alpha \in k$.

Pf: We know

$$f(x) \mid x^{p^d} - x \in \mathbb{F}_p[x].$$

We also know that every element of k is a zero for $x^{p^d} - x$.

Write $x^{p^d} - x = f(x) \underbrace{g(x)}_{\deg = p^d - r} \in \mathbb{F}_p[x]$

So $g(x)$ has at most $p^d - r$ zeros in k .

↳ But $x^{p^d} - x$ has p^d zeros in k

$\Rightarrow f(x)$ must have r distinct zeros in k .

Rmk: We have shown:

$f(x) \in \mathbb{F}_q[x]$ irred. of deg r
 $|k| = p^d$

TFAE:

(1) $r \mid d$

(2) $f(x)$ has r distinct zeros in k

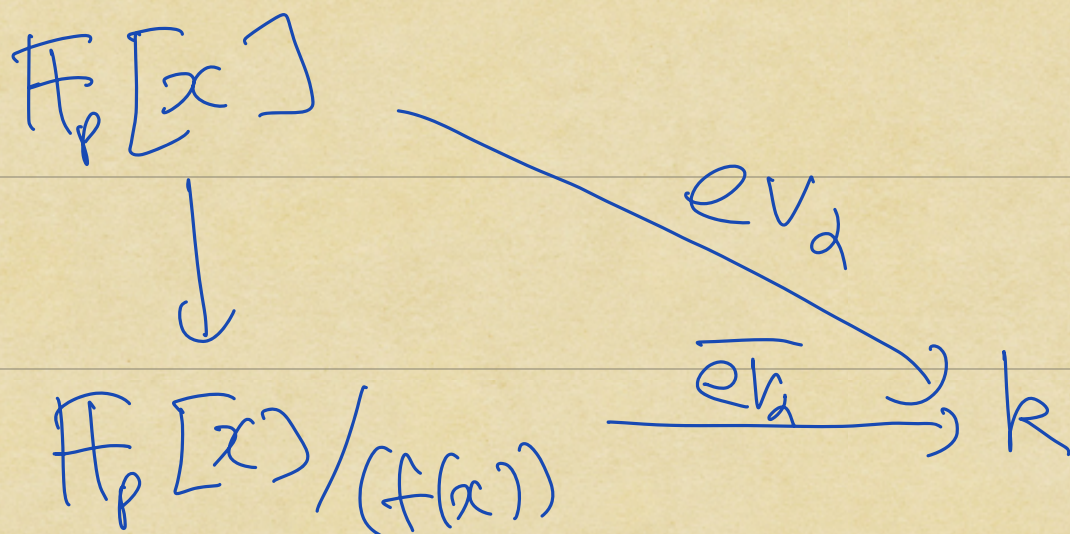
(3) $f(x)$ has a zero in k

Pf: (1) $\Rightarrow$ (2) (Proved in proposition)

(2) $\Rightarrow$ (3) ✓

(3) $\Rightarrow$ (1)

$\alpha \in k$ is a zero for $f(x)$



$\bar{\text{ev}}_\alpha$ is injective

$\Rightarrow \mathbb{F}_p[x]/(f(x)) \xrightarrow{\bar{\text{ev}}_\alpha} \text{im } \bar{\text{ev}}_\alpha \leq k$
 $\uparrow$
sub-field

But $|\text{im } \bar{\text{ev}}_\alpha| = |\mathbb{F}_p[x]/(f(x))| = p^r$

$$\& \quad |k| = p^d$$

$$\Rightarrow r \mid d.$$

HW 3

Problem 4(b)

Defⁿ $f(x) \in R[x]$ (R : comm. ring)

$$f(x) = \sum_{i=0}^m a_i x^i$$

Then the derivative $f'(x)$
is the polynomial

$$f'(x) = \sum_{i=1}^m i a_i x^{i-1}$$

Fact (3)

$$\bullet (f(x) + g(x))' = f'(x) + g'(x)$$

$$\bullet (cf(x))' = c f'(x)$$

Fact (4)

$$(\underline{f(x)}g(x))' = f'(x)g(x) + f(x)g'(x)$$

Pf: Both sides are separately
linear in $f(x)$ & $g(x)$

So Fact (4) can be reduced to
the case where $f(x) = x^m$
 $g(x) = x^n$

$$(x^m \cdot x^n)' \stackrel{?}{=} (x^m)'x^n + x^m \cdot (x^n)'$$

$$\parallel \qquad \parallel$$

$$(n+m)x^{n+m-1} \stackrel{\checkmark}{=} mx^{m-1}x^n + x^m \cdot nx^{n-1}$$

Proposition: k : field

$$0 \neq f(x) \in k[x]$$

Then: $f(x)$ has repeated irreducible factors $\Leftrightarrow \gcd(f(x), f'(x)) \neq 1$

Pf: $f(x) = c \cdot q_1(x)^{d_1} q_2(x)^{d_2} \dots q_r(x)^{d_r}$

where $q_1(x), q_2(x), \dots, q_r(x)$
are distinct irred. poly. with
leading coeff 1

If $d_1 \geq 2$, then

$$f(x) = q_1(x)^2 \cdot h(x)$$

$$\Rightarrow f'(x) = 2q_1(x)q_1'(x) \cdot h(x) + q_1(x)^2 \cdot h'(x)$$

$$\in (q_1(x))$$

$$\Rightarrow \gcd(f(x), f'(x)) \neq 1$$

For the converse, suppose that

$$\gcd(f(x), f'(x)) = h(x) \text{ has } \deg \geq 1$$

We can assume that $q_1(x)$ is an irreducible factor of $h(x)$.

$$f(x) = q_1(x)^{d_1} g(x)$$

$$f'(x) = d_1 q_1(x)^{d_1-1} g(x) + q_1(x)^{d_1} g'(x)$$

The only way for $q_1(x)$ to divide the RHS is to have $d_1 \geq 2$

i.e. $f(x)$ must have repeated factors.

Example

$$x^p \in \mathbb{F}_p[x]$$

$$(x^p)' = p \cdot x^{p-1} = 0!!$$